Dilnath Rathnayaka

Software Engineering Undergraduate

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Matara,SriLanka

PROFESSIONAL SUMMARY

Motivated 2nd-year BEng Software Engineering student at IIT. Skilled in object-oriented programming and backend frameworks including Go (Gin), Java, and Python. Experienced with MySQL, MariaDB, AWS RDS, and Docker. Built 'QuickRead', an OCR-based prescription scanner, proving my ability to deliver scalable solutions. Seeking a software engineering internship to apply my technical abilities

EDUCATION

BEng.(Hons) Software Engineering | University Of Westminster affiliated with Informatic institute of technology

Secondary Education | St.Thomas College Matara

PROJECTS

Quick Read - Smart Prescription Scanning and Pharmacy Delivery System

Developed the backend architecture for an innovative healthcare platform designed to streamline the pharmacy experience through automated data extraction and real-time delivery tracking.

- **Backend Architecture:** Engineered a robust, RESTful API using Go (Gin) to handle secure user authentication, role-based access control, and order processing.
- **Database Engineering:** Designed and implemented a relational database schema using MySQL (hosted on AWS RDS) to manage patient data, inventory, and dynamic order statuses.
- **Smart Scanning Integration:** Implemented endpoints to handle image uploads and integrated OCR/NLP technologies to extract and process text from physical prescription images.
- **Real-Time GPS Tracking:** Developed a dynamic location-tracking module allowing delivery drivers to update their coordinates via the API for real-time customer tracking.
- **Containerization & Deployment:** Utilized Docker to maintain consistent database and application environments during development.
- **Technologies:** Go, Gin, Python (Flask), React Native, MySQL, AWS RDS, Docker, REST APIs, OCR/NLP, JSON Web Tokens (JWT), HTML/CSS, Git.

Ultra Engineering – Corporate Web Platform & Industrial Portfolio

Role: Lead Full-stack Developer

- **Project Overview:** Engineered and deployed a comprehensive corporate web platform for a leading engineering firm in Sri Lanka specializing in high-voltage diagnostics and cable fault scanning.
- **Technical Implementation:** Developed a high-performance, responsive UI using **React.js** and **Tailwind CSS**, focusing on optimized user experience and fast load times via **Vite**.
- **Portfolio Digitalization:** Successfully transitioned extensive offline project records (including major national projects like the **Kelanithissa Powerplant** and **southern Expressway**) into a dynamic digital showcase.
- **SEO & Analytics:** Implemented advanced **SEO strategies** including custom Meta tags, XML Sitemaps, and **Google Search Console** integration, achieving organic visibility for niche industrial keywords.
- **System Integration:** Integrated an automated lead generation system using **Web3Forms** and configured custom **DNS/SSL** via Netlify for secure, professional business operations.
- **Technologies:** React.js, Tailwind CSS, Vite, Netlify, Web3Forms

Zero Hunger Initiative Website

Collaborated on the design and development of a responsive website dedicated to promoting the UN's Sustainable Development Goal 2: Zero Hunger.

- **Front-End Development:** Architected and developed the 'User Profile' and 'Sitemap' pages from scratch using modern web standards.
- **Responsive Design:** Implemented mobile-first design principles to ensure seamless functionality and visual consistency across desktops, tablets, and smartphones.
- **User Experience (UX):** Designed an intuitive Sitemap to enhance website navigation and a dynamic User Profile interface to drive user engagement.
- **Problem Solving:** Debugged front-end logic and refined CSS styling to ensure cross-browser compatibility.
- **Technologies:** HTML5, CSS3, JavaScript, GitHub.

European Airports Data Analysis System

Developed a standalone Python application to process thousands of flight records and analyze European air traffic patterns, weather impacts, and airline performance.

- **Data Processing:** Engineered an algorithm to parse CSV datasets into nested Python lists, extracting complex metrics such as delayed flight percentages and average hourly departures.
- **Input Validation:** Built a robust, interactive command-line interface with strict data validation to ensure user inputs met exact formatting and range requirements.
- **Data Visualization:** Designed and implemented an auto-scaling histogram using a custom graphics module to visually represent hourly flight departures for specific airlines.
- **File Handling & Architecture:** Created a continuous-loop program allowing users to process multiple datasets in one session, automatically appending results to a persistent text file.
- **Technologies:** Python, Tkinter.

SKILLS

- **Languages:** Go (Golang), Python, Java, JavaScript, HTML5, CSS3
- **Databases & Cloud:** MySQL, MariaDB, AWS RDS, Microsoft Azure AI Studio
- **Frameworks & Tools:** React.js, React Native, Angular.js, Flask, Docker, Git Core
- **Concepts:** REST APIs, JSON Web Token (JWT), Data Analysis, OCR, NLP, UI/UX

CERTIFICATIONS AND ACHIEVEMENTS

- **Azure Ai Essentials** | Microsoft | 2025-07
- **CSS skill certification** | HackerRank | 2025-07
- All Island Music Competition | 2nd place winner, awarded Dhrupad vocal and Dhrupad instrumental (2018, 2019)
- Leadership| College Prefect (2018–2019), assisted in maintaining school discipline and supporting events

FIND ME ONLINE

Github - https://github.com/Dilnath225 Linked-in - www.linkedin.com/in/dilnathrathnayaka-3282a127b

LANGUAGES

- Sinhala
- English

REFERENCES

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